

Personal Library & AI Research Engine — Domain Entity Model & Database ERD

Comprehensive Database Schemas: MongoDB BSON Documents • Qdrant Vector Points • Keycloak Security Sub • Version Snapshot Lineage

Entity Schema v1.1.0

0..1

1 : 2 (Dual)

1 : N

1 : N

1 : 1

1 : N

1:N

1 : N

1 : N

DocumentEntity (Aggregate Root) MongoDB [documents]

```
* guid : UUID <<PK, Indexed>>
previousVersionGuid : UUID? <<Self-Ref FK>>
versionNumber : Integer (starts at 1)
fileName : String (e.g. attention.pdf)
fileSize : Long (bytes)
fileSizeFormatted : String (e.g. 2.4 MB)
format : String (pdf, docx, txt, md, pptx)
uploadDate : Instant (ISO-8601)
editDate : Instant (latest update)
contentExcerpt : String (first 300 chars)
fullContent : Text (extracted plain text)
chunkCount : Integer (partitioned count)
tags : List<String> (user defined)
sha256Hash : String (integrity digest)
uploadedBy : String <<FK -> UserProfile.id>>
```

1:1 Composition: bibtex (BibTeXMetadata)
1:2 Composition: summaries (Llama, Mistral)
1:N Partitions: chunks (DocumentChunk[])
1:N Versioning: versionHistory (Snapshots[])
0..1:1 Self-Ref: previousVersionGuid (Lineage)

BibTeXMetadata (Embedded) Value Object

```
* bibKey : String <<Synthesized PK>>
entryType : BibTeXType (article, book...)
title : String (publication title)
author : String (canonical author list)
year : String (e.g. '2017')
month : String? (e.g. 'December')
journal : String? (academic journal)
booktitle : String? (proceedings)
volume : String? / number : String?
pages : String? (e.g. '5998-6008')
publisher : String? (publishing house)
edition : String? (e.g. '1st Edition')
institution : String? / school : String?
doi : String? <<DOI Lookup Index>>
isbn : String? / issn : String?
url : String? (canonical link)
abstract : Text (academic summary)
keywords : String (comma tags)
rawBibTeX : Text (LaTeX syntax)
```

AST Lexer generated from document text

SummaryRecord (Dual-Model) Ollama AI

```
* modelKey : 'llama' | 'mistral' <<PK>>
modelName : String (Llama 3.3 / Mistral)
summaryText : Text (Markdown format)
generatedAt : Instant (timestamp)
durationSeconds : Double (inference time)
durationFormatted : String ('14.2s')
promptTokens : Integer (input length)
completionTokens : Integer (output)
temperature : Double (0.2 Llama, 0.3 Mistral)
```

Eliminates single-model cognitive bias

DocumentChunk (Vector) Qdrant DB

```
* id : String <<PK: guid-c{idx}>>
* documentGuid : UUID <<FK, Payload Filter>>
chunkIndex : Integer (0, 1, 2...)
text : Text (~400-500 tokens)
pageNumber : Integer (source page)
tokenCount : Integer (sub-words)
denseVector : float32[768] <<Vector>>
hnswIndex : HNSW (Cosine, M=16, ef=100)
```

Collection: personal_library_embeddings

RAGConversation Session Q&A

```
* conversationId : UUID <<PK>>
* documentGuid : UUID <<FK -> DocumentEntity>>
userQuery : String (natural language query)
answerText : Text (synthesized RAG answer)
timestamp : Instant (query timestamp)
```

1:N Citations: references (CitationReference[])
Grounds generation in retrieved vector context

CitationReference (Grounding) Evidence

```
* id : UUID <<PK>>
* conversationId : UUID <<FK -> Conversation>>
* chunkId : String <<FK -> DocumentChunk.id>>
citationIndex : Integer ([1], [2] badge)
similarityScore : Float (e.g. 0.918 = 91.8%)
sourceSnippet : Text (verbatim quote)
pageNumber : Integer / charOffset : Integer
```

Powers interactive source inspection drawer

DocumentVersionSnapshot (Audit) Immutable Chain

```
* snapshotGuid : UUID <<PK, Historical ID>>
* documentGuid : UUID <<FK -> DocumentEntity>>
versionNumber : Integer (archived revision)
fileName : String (e.g. attention_v1.pdf)
fileSize : Long / fileSizeFormatted : String
format : String (pdf, docx, etc.)
savedAt : Instant (snapshot timestamp)
sha256Hash : String (immutable digest)
bibtexRaw : Text (historical BibTeX entry)
llamaSummary : Text? / mistralSummary : Text?
contentExcerpt : String / fullContent : Text?
chunksCount : Integer (chunks at snapshot)
changeLogNote : String (audit description)
modifiedBy : String <<FK -> UserProfile.id>>
```

Non-Destructive Rollback Target

Restoring version N creates version M = max(v) + 1 without purge

UserProfile (Keycloak IAM) OIDC JWT

```
* id : UUID <<PK, Keycloak Sub Claim>>
username : String <<Unique Login>>
email : String (verified corporate email)
firstName : String / lastName : String
realm : String ('personal-library-realm')
roles : Set<String> <<RBAC Claims>>
```

RBAC Roles: LIBRARY_ADMIN, CHIEF_RESEARCHER

Token: RS256 signed JSON Web Token (RFC 7519)

CatalogFilterState & Query Projection Model

Analytical List Report

```
fileName : String? (regex match)
title : String? (case-insensitive fuzzy)
author : String? (author filter)
edition : String? / format : String? (enum)
content : String? (Qdrant semantic search)
page : Integer (1-based) / pageSize : Integer (10, 20)
sortBy : fileName | title | author | edition | uploadDate
sortOrder : asc | desc
```

Reactive Store: FilterStore (TypeScript) <--> Spring Data MongoDB Criteria Builder
Binds UI5 FilterBar inputs to backend paginated REST queries (/api/v1/documents)